

ROCK HOUND

DESCRIPTION: Teams will demonstrate knowledge of types and properties of rocks and minerals as well as natural resources.

COMPETITORS BRING:

- Writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

NUMBER OF PARTICIPANTS: Maximum of 2

IMPOUND: NO

THE COMPETITION:

1. The event may be run as stations, may include a slideshow of images, and/or may be a “pencil and paper” test.
2. Teams will identify as many rocks and minerals as possible from a selected group (may be from images or samples). The list of rocks and minerals will come from the following list:

ROCKS:	basalt granite pumice scoria	bituminous coal limestone (fossil) quartzite shale	conglomerate marble sandstone slate	gneiss obsidian schist (garnet) limestone (chemical)
MINERALS:	calcite galena Hematite Quartz (chert)	copper graphite mica-biotite quartz (crystal)	feldspar (pink) gypsum-satin-spar pyrite talc	fluorite halite kaolinite

3. No more than half of the questions on the exam will be identification questions.
4. Additional questions may be from any of the following topics:
 - a. Categories of rocks (igneous, metamorphic, sedimentary)
 - b. The rock cycle
 - c. Weathering and erosion
 - d. Characteristics of minerals (hardness, luster, color, cleavage, streak)
 - e. Natural resources (renewable vs. nonrenewable, conservation of resources, Florida resources such as phosphate and limestone)
5. Students may be asked to perform a streak test.

RELATED FLORIDA STANDARDS:

SC.4.E.6.1 Identify the three categories of rocks: igneous, (formed from molten rock); sedimentary (pieces of other rocks and fossilized organisms); and metamorphic (formed from heat and pressure).

SC.4.E.6.2 Identify the physical properties of common earth forming minerals, including hardness, color, luster, cleavage, and streak color, and recognize the role of minerals in the formation of rocks.

SC.4.E.6.3 Recognize that humans need resources found on Earth and that these are either renewable or nonrenewable.

SC.4.E.6.6 Identify resources available in Florida (water, phosphate, oil, limestone, silicon, wind, and solar energy).

SCORING:

1. High score wins.
2. Each question or station will be assigned a predetermined set of points, assigned by the event supervisor and communicated to the teams (preferably- points are identified on the test/station paperwork).
3. Ties will be broken with pre-determined tie-breaker questions which are communicated to teams.